

Understanding Analysis, 2nd Edition

Chapter 6 - Section 6.4 - Series of Functions

1. Supply the details for the proof of the Weierstrass M-Test (Corollary 6.4.5).

Solution: We have for every fixed x

$$0 \leq |f_n(x)| \leq M_n$$

and $\sum_{n=1}^{\infty} M_n$ converges. This satisfies the conditions for the Comparison Test, so $\sum_{n=1}^{\infty} |f_n(x)|$ converges.

Then by the Absolute Convergence Test, $\sum_{n=1}^{\infty} |f_n(x)|$ converges implies $\sum_{n=1}^{\infty} f_n(x)$ converges.

It remains to prove that the convergence is uniform. $\sum_{n=1}^{\infty} M_n$ converges so it satisfies the Cauchy Criterion for Series (Theorem 2.7.2): for any $\epsilon > 0$, there exists N such that for all $n > m \geq N$:

$$M_{m+1} + M_{m+2} + \dots + M_n < \epsilon$$

Reusing the same N , we can prove that $\sum_{n=1}^{\infty} f_n(x)$ satisfies the Cauchy Criterion for Uniform Convergence of Series (Theorem 6.4.4). Because $|f_n(x)| \leq M_n$, then

$$\begin{aligned} |f_{m+1}(x) + f_{m+2}(x) + \dots + f_n(x)| &\leq |f_{m+1}(x)| + |f_{m+2}(x)| + \dots + |f_n(x)| \\ &\leq M_{m+1} + M_{m+2} + \dots + M_n \\ &< \epsilon \end{aligned}$$

Since the same N works for any x (because the bound M_n is independent of x), the convergence is uniform.

2. Short proof or counterexample.

- (a) If $\sum_{n=1}^{\infty} g_n$ converges uniformly, then (g_n) converges uniformly to zero.
- (b) If $0 \leq f_n(x) \leq g_n(x)$ and $\sum_{n=1}^{\infty} g_n$ converges uniformly, then $\sum_{n=1}^{\infty} f_n$ converges uniformly.
- (c) If $\sum_{n=1}^{\infty} f_n$ converges uniformly on A , then there exist constants M_n such that $|f_n(x)| \leq M_n$ for all $x \in A$ and $\sum_{n=1}^{\infty} M_n$ converges.

Solution:

- (a) True. This is analogous with the n th-term test for series of numbers (Theorem 2.7.3).

$\sum_{n=1}^{\infty} g_n$ converges uniformly so there exists N such that for all $n > m > N$ and for all x

$$|g_{m+1}(x) + \dots + g_n(x)| < \epsilon$$

The same N works to prove (g_n) converges uniformly. Set $n = m + 1$.

Alternatively, without using the Cauchy Criterion, let $s_k = \sum_{n=1}^k g_n(x)$. Then

$$\lim_{k \rightarrow \infty} s_k - s_{k-1} = \lim_{k \rightarrow \infty} s_k - \lim_{k \rightarrow \infty} s_{k-1} = 0$$

And note that $s_k - s_{k-1} = g_k$, so $\lim_{k \rightarrow \infty} s_k - s_{k-1} = \lim_{k \rightarrow \infty} g_k(x) = 0$.

The important intuition here is that uniform convergence allows us to reuse many results from series of numbers, because any bound works uniformly so we may treat $g_n(x)$ for all x like a single entity.

- (b) True. This is analogous to the Comparison Test for series of numbers (Theorem 2.7.4).

Like part (a), but we can get rid of the absolute values because it's all non-negative

$$g_{m+1}(x) + \dots + g_n(x) < \epsilon$$

then reusing the same N and the problem's assumption that $0 \leq f_n(x) \leq g_n(x)$

$$f_{m+1}(x) + \dots + f_n(x) \leq g_{m+1}(x) + \dots + g_n(x) < \epsilon$$

- (c) False. This is analogous to conditional convergence for series of numbers (Theorem 2.7.8).

Take $f_n(x) = \frac{(-1)^{n+1}}{n}$ for all x . Then $\sum_{n=1}^{\infty} f_n$ is the alternating harmonic series for all x so it converges uniformly.

But $M_n \geq \frac{1}{n}$ so $\sum_{n=1}^{\infty} M_n$ diverges by comparison test with $\sum_{n=1}^{\infty} \frac{1}{n}$.

3. (a) Show that

$$g(x) = \sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}$$

is continuous on all of $\mathbf{R}$.

- (b) The function g was cited in Section 5.4 as an example of a continuous nowhere differentiable function. What happens if we try to use Theorem 6.4.3 to explore whether g is differentiable?

Solution:

- (a) We have $|\cos(2^n x)| \leq 1$ for any n and x . Then

$$\left| \frac{\cos(2^n x)}{2^n} \right| \leq \frac{1}{2^n}$$

for all x and $\sum_{n=0}^{\infty} \frac{1}{2^n} = 2$. This satisfies the Weierstrass M-test, so $\sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}$ converges uniformly.

By Theorem 6.4.2 (Term-by-term Continuity Theorem), since $\frac{\cos(2^n x)}{2^n}$ is continuous for any n and $\sum_{n=0}^{\infty} \frac{\cos(2^n x)}{2^n}$ converges uniformly, then g is continuous.

- (b) The series of derivative is

$$\sum_{n=0}^{\infty} -\sin(2^n x)$$

This fails to converge uniformly so we can't apply Theorem 6.4.3. For example, take $x = \pi/3$, then the terms are

$$-\sin\left(\frac{2^n \pi}{3}\right) = \begin{cases} -\frac{\sqrt{3}}{2}, & n = 0 \\ (-1)^n \frac{\sqrt{3}}{2}, & n \geq 1 \end{cases}$$

Since the terms fail to approach 0, the series fails to converge.

This does not prove that g is not differentiable; it only means we can't use Theorem 6.4.3 to conclude differentiability.